

Nguyen Van Kinh

AI Engineer

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OBJECTIVE

Final-year Data Science undergraduate (GPA 3.81/4) at the University of Science – VNU-HCM, currently a Junior AI Engineer. Passionate about turning research ideas into reliable, production-grade AI products. Strengths in LLM-powered systems — RAG, multi-agent architectures, and knowledge graphs over unstructured data — with a strong emphasis on evaluation, monitoring, and continuous improvement. Experienced in shipping enterprise conversational AI and data-intelligence platforms end to end.

EDUCATION

University of Science, Vietnam National University Ho Chi Minh City	2022 – 2026
Major: Data Science	GPA: 3.81/4

WORK EXPERIENCE

Junior AI Engineer <i>Namitech</i>	Feb 2026 – Present
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Project: NamiGen – Multi-tenant Conversational AI Platform (Voicebot/Chatbot)

Platform engineer on a production conversational AI platform powering voice and chat assistants for 10–15 concurrent enterprise tenants (prod/PoC/demo) from a single, fully config-driven codebase — dialog flows, prompts, knowledge bases, and wire protocols are declared per customer, so new PoCs ship without touching engine code.

Responsibilities:

- Built the graph-based dialog engine (speak/listen state machine) — LLM intent routing, global actions (repeat, go-back, pause, human hand-off), silence policies, and flow navigation robust to branches, loops, and session restores — serving intent-only script turns at **≈3s** p50 latency.
- Built RAG answering pods with hybrid retrieval (Qdrant) and strict-fidelity prompting reaching **Recall@5 ≈ 78%** and **Precision@5 ≈ 70%**, with an **85% Faithfulness** score (DeepEval) guarding against hallucinated facts; full RAG turns (intent + retrieval + generation) served at **≈5.5s** p50 across Vietnamese, Japanese, and English deployments.
- Designed real-time WebSocket delivery with pluggable per-customer wire protocols (streaming chat, step-synced voice clients) and partner-facing REST APIs secured with API-key / JWT authentication.
- Extended the multi-tenant configuration architecture (customer/scenario/ pod layers, policies, taxonomies) so new PoCs ship in **1–3 weeks** instead of the prior **1–2 months**, keeping reusable logic in the engine and customer-specific behavior purely in config.

Selected PoCs shipped on the platform:

- **Rinnai (Japan):** Japanese voice cooking assistant, iPad client — slide-synced, zero-hallucination FAQ.
- **AI Interview (Recruiter):** automated STAR-scoring API using LLM-as-a-judge with weighted rubrics.
- **BIDV:** banking call-center voicebot for inbound support and outbound campaigns.
- **PVI:** insurance support bot with automated ticket-creation pipeline.
- **Unilever:** in-store retail voice assistant for point-of-sale product guidance.
- **Namiweb:** RAG-based website chat assistant for lead engagement.

Tech Stack: Python, FastAPI, WebSocket, MongoDB, Qdrant, Redis, Docker, Langfuse, DeepEval, Azure-OpenAI

AI Engineer – Internship

Jul 2025 – Jan 2026

MetaData Solutions

Project: 1Brain – Data Intelligence Platform

Contributed to a multi-agent data intelligence platform that transforms unstructured documents into structured graph data, enabling advanced querying, exploration, and analytical visualization.

Responsibilities:

- Designed and implemented a multi-agent architecture with graph-based retrieval that **improved query accuracy from 66% to 85%** versus native RAG, processing **10–20 concurrent documents** (15–30

pages each) into structured knowledge graphs for complex querying and reasoning.

- Developed analytical visualization components to support relationship exploration and data-driven insights.

Tech Stack: LangChain, LangGraph, LangSmith, FastAPI, FalkorDB, PostgreSQL, RAG.

Project: Telehealth – Chatbot Medical Assistant for Doctors

Developed a knowledge graph-powered medical assistant capable of transforming unstructured medical records into structured insights and performing autonomous administrative actions.

Responsibilities:

- **Improved chatbot accuracy from 62% to 83% (F1-score)** by grounding responses in clinically verified graph data, supporting complex medical queries.
- Developed a custom toolset enabling the AI agent to execute CRUD operations on patient databases, facilitating real-time record updates and autonomous appointment scheduling.
- Engineered a multi-stage pipeline (via Graphiti) transforming unstructured medical Google Docs (text and images) into a structured patient-centric knowledge graph using FalkorDB.

Tech Stack: LangChain, Graphiti, FalkorDB, FastAPI, LangFuse, RAG.

PERSONAL PROJECTS

Multi-Agent System for Multimedia Soccer Knowledge Understanding | [GitHub](#)

Role: AI Engineer / Core Developer

Multi-agent conversational AI ecosystem embedded into online soccer streaming platforms, enabling real-time news/web search and live in-broadcast Q&A during matches, by orchestrating structured, vector, and cache stores with dynamic multi-worker planning and async background media processing.

Key Features & Architecture:

- **Graph-Based Multi-Agent Orchestration:** **FastAPI** + **LangGraph** with dynamic DAG routing, strict node separation, and PostgreSQL checkpointing for multi-turn persistence and thread-safe crash recovery.
- **Multi-Intent Planning & Disambiguation:** **LiteLLM**-routed planning node resolving co-references/abbreviations via top-3 few-shot templates from a **Qdrant** CaseBank, decomposing multi-intent questions into parallel tool-call chains in one pass.
- **Multi-Tier Memory & Cognitive Compression:** memory pipeline filtering raw tool traces, semantic deduplication via cosine similarity (≥ 0.95), and token-aware context pruning to prevent context-window overflow.
- **Asynchronous Live-Video Indexing:** decoupled **Celery** + **Redis** worker intercepting live HLS segments, extracting audio via FFmpeg and transcribing through a rolling-window **Qwen3-ASR** pipeline into MongoDB.
- **Multi-Modal Facial Search:** **Qwen-VL** bounding-box grounding + **InsightFace (ArcFace)** 512-D embeddings, served via Qdrant over **gRPC** for 1-to-N player identification.

Key Performance Outcomes:

- **SoccerBench State-of-the-Art (SOTA):** Elevated average cross-task accuracy from **69.3%** to **75.5%** (peak **96.4%** on domain text QA), outperforming the base multi-modal soccer agent benchmark.
- **Low-Latency Semantic Caching:** Cut repeated-query response time by **55%** (12.1s $\rightarrow$ 5.5s p50) via Redis-backed semantic caching, bypassing tool execution on cache hits.
- **Near-Constant-Time Parallel Scaling:** $T_{worker} \approx \max(T_i)$ — a 3-entity parallel image query (21.2s p50) added only **0.6s** over a single-entity query (20.6s p50), versus $\sim 3\times$ for a naive sequential design.

Technical Stack: FastAPI, LangGraph, LangChain, Langfuse, Google Gemini, Qwen, PostgreSQL, MongoDB, Qdrant, Redis, OpenCV, FFmpeg, DeepFace.

AWARDS & ACHIEVEMENTS

- **AI Challenge TP.HCM 2026** — Building an intelligent multimodal video search & retrieval system; responsible for the search core engine (FAISS HNSW+SQ8 over $\sim 2\text{M}$ vectors, two-tier recall $\rightarrow$ rerank pipeline, gRPC service).

SKILLS

- **Generative AI & LLMs:** LangChain, LangGraph, LangSmith, LangFuse, DeepEval, Graphiti, HuggingFace, RAG Pipelines, Multi-agent Systems.
- **Deep Learning & CV:** PyTorch, TensorFlow, Scikit-learn, OpenCV, OCR, Object Detection.
- **Backend & Web:** FastAPI, Flask, Next.js, OpenAPI, Python.
- **Databases & Infra:** FalkorDB, Qdrant, PostgreSQL, MongoDB, AWS S3, Docker, Git.